ROADRESQ

Product Requirements Document

Version 1.0 • Product Definition • India

“When your vehicle breaks down, RoadResQ gets the right help to you.”

Roadside Assistance & Automotive Service Marketplace

Master product specification for RoadResQ.

1. Executive Summary

RoadResQ is a digital roadside assistance and automotive service platform connecting vehicle owners with nearby verified mechanics, garages, towing providers, battery technicians, tyre technicians, and other automotive professionals.

The platform manages the complete journey: Request → Match → Track → Inspect → Approve → Repair → Pay → Review. Providers receive tools for jobs, customers, vehicles, estimates, invoices, availability, and earnings.

2. Problem Statement

Vehicle breakdowns are stressful because roadside assistance is fragmented. Customers often search manually, call several providers, explain the problem repeatedly, share their location, negotiate pricing, and wait without reliable visibility.

Small and medium garages also commonly depend on phone calls, WhatsApp, paper registers, spreadsheets, manual job cards, and informal records.

- Slow discovery of suitable assistance
- Uncertainty about availability and ETA
- Limited trust in unknown providers
- Unclear pricing and approvals
- Poor visibility during active jobs
- Fragmented service history
- Manual garage operations and billing

3. Product Vision

Build a trusted digital infrastructure for vehicle assistance and automotive services.

RoadResQ should become the platform vehicle owners think of whenever they need emergency assistance or automotive service—not merely a mechanic directory.

4. Product Mission

- Make assistance fast and easy to request
- Connect customers with verified providers
- Improve pricing and service transparency
- Provide real-time job visibility
- Maintain digital vehicle and service records
- Give mechanics and garages useful digital tools

5. Target Users

User	Purpose
Customer	Requests emergency assistance, towing, repairs, or scheduled servicing.
Service Provider	Mechanic, garage, towing operator, battery/tyre technician, or other automotive professional.

Administrator	Manages verification, users, bookings, disputes, payments, support, and analytics.

6. Core Services

- Battery assistance / jump-start
- Flat tyre and puncture assistance
- Vehicle towing and recovery
- Emergency fuel assistance
- Vehicle-won't-start assistance
- Mechanical roadside repair
- Accident assistance and recovery
- General roadside assistance
- Scheduled garage service

7. Core Product Loop

```
REQUEST
↓
MATCH
↓
ACCEPT
↓
TRACK
↓
ARRIVE
↓
INSPECT
↓
APPROVE
↓
REPAIR
↓
PAY
↓
REVIEW
```

8. Customer Experience

Home: Primary action is GET HELP NOW, supported by Find Garage, Book Service, My Vehicles, and Service History.

Emergency: Select vehicle → select problem → confirm location → optionally add description/media → request help.

Matching: Rank providers by distance, capability, availability, vehicle compatibility, rating, ETA, and performance.

Completion: Provider verifies arrival, inspects, estimates when needed, completes the job, invoices, and receives payment/review.

9. Emergency Assistance Requirements

- Request assistance with minimal friction
- Select or add a vehicle
- Choose a problem category

- Confirm GPS location and adjust it manually
- Attach optional photos/video
- Search only eligible available providers
- Create an active booking after acceptance
- Send clear status updates

10. Provider Matching Engine

```
Candidate Providers
↓
Service Capability Filter
↓
Vehicle Compatibility Filter
↓
Availability Filter
↓
Distance / ETA Calculation
↓
Rating + Performance Ranking
↓
Request Dispatch
```

If a provider declines or does not respond within the configured timeout, the system should attempt another eligible provider.

11. Live Tracking & Job Status

During an active job, show provider location, route/ETA where available, provider identity, service type, contact options, and current status.

```
REQUESTED → SEARCHING → PROVIDER_ASSIGNED → ACCEPTED
→ ON_THE_WAY → ARRIVED → INSPECTION → ESTIMATE_PENDING
→ APPROVED → IN_PROGRESS → COMPLETED
→ INVOICE_GENERATED → PAYMENT_COMPLETED → CLOSED
```

12. Arrival Verification

An OTP-based arrival verification can confirm that the correct provider has reached the customer and transition the job into ARRIVED.

13. Inspection, Estimates & Approval

Providers can create digital inspections containing diagnosis, notes, photos, recommended services, parts, labour, and estimated costs.

Additional or complex work should be submitted as an estimate and require customer approval before proceeding.

14. Pricing

Show estimated or fixed pricing when it can be determined reliably. For complex repairs, clearly state that final pricing requires inspection.

```
Base Service
+ Travel / Distance
+ Parts
+ Labour
+ Applicable Platform Fee
= Customer Total
```

15. Digital Job Card

Each service creates a digital job record containing job ID, customer, vehicle, reported problem, provider, diagnosis, services performed, parts, labour, status, and timestamps.

16. Invoices & Payments

Completed jobs produce digital invoices with itemized charges and payment status. MVP can prioritize UPI and cash, with cards and other methods later.

- View invoice
- Download/share invoice
- Record payment status
- Retry failed payments
- Maintain transaction history

17. Vehicle Management

Customers can store multiple vehicles with manufacturer, model, variant, year, registration number, fuel type, and service history.

```
MY VEHICLES
├── Vehicle A
├── Vehicle B
└── + ADD VEHICLE
```

18. Service History

Completed services are linked to the vehicle and shown as a chronological timeline, creating long-term value and enabling future maintenance reminders.

19. Garage Discovery & Scheduled Service

Customers can search nearby garages using distance, rating, opening status, service type, vehicle compatibility, verification, and pickup/drop availability.

Flow: Select Service → Select Garage → Select Date → Select Time → Confirm Booking.

20. Provider Application

- Registration and verification
- Business/profile information
- Services and supported vehicle types
- Online/offline availability
- Service radius and working hours
- Incoming request notifications
- Accept/reject requests
- Job status updates
- Inspection and estimate creation
- Job completion
- Earnings dashboard

21. Garage Management

For larger providers, RoadResQ can provide customers, vehicles, job cards, mechanics, inventory, estimates, invoices, payments, and reports. This layer can later evolve into the broader Better Garage product.

22. Admin Platform

- Dashboard and operational overview
- Customer management
- Provider management and verification
- Active/completed/cancelled job monitoring
- Payment and payout monitoring
- Support tickets and disputes
- Fraud/safety monitoring
- Platform analytics

23. Provider Verification

```
REGISTER
↓
BUSINESS DETAILS
↓
IDENTITY / ADDRESS
↓
DOCUMENTS
↓
SERVICE CAPABILITY
↓
ADMIN REVIEW
↓
VERIFIED
↓
ACTIVE
```

Provider states: PENDING, VERIFIED, ACTIVE, SUSPENDED, REJECTED.

24. Notifications & Communication

- Customer: request, provider found, acceptance, arrival, estimate, completion, invoice, payment, review
- Provider: new request, cancellation, location update, estimate approval, payment
- In-app communication: chat, masked calling, and quick messages

25. Safety & Support

RoadResQ is a vehicle-assistance platform and must clearly distinguish vehicle assistance from public emergency services. It may provide emergency-contact/location-sharing shortcuts and RoadResQ support.

Customers can report no-show, pricing disputes, poor service, incorrect invoices, payment issues, or inappropriate behaviour.

26. Roles & Permissions

Role	Primary Access
Super Admin	Full platform control
Operations Admin	Bookings, providers, support

Verification Admin	Provider verification
Finance Admin	Payments, invoices, payouts
Provider	Own jobs, customers, services, earnings
Mechanic	Assigned jobs and service details
Customer	Own vehicles, bookings, payments, history

27. Database Architecture

```
users
vehicles
providers
provider_documents
provider_services
provider_availability
services
bookings
booking_locations
booking_status_history
job_cards
inspections
inspection_items
estimates
estimate_items
invoices
invoice_items
payments
reviews
notifications
messages
support_tickets
payouts
audit_logs
```

Core relationship: User → Vehicle → Booking → Provider → Job Card → Estimate → Invoice → Payment → Review.

28. Recommended Technology Stack

- Frontend: Next.js, React, TypeScript, Tailwind CSS
- UI/Animation: Framer Motion; GSAP where appropriate
- Backend: FastAPI / Python
- Database: PostgreSQL + PostGIS
- Real-time: WebSockets + Redis
- Maps: Google Maps or Mapbox
- Authentication: OTP/email + role-based authorization
- Object storage: vehicle images, provider documents, inspection media, invoices

29. High-Level Architecture

```
CUSTOMER / PROVIDER CLIENTS
↓
API LAYER
↓
ROADRESQ BACKEND
↙ ↓ ↘
AUTH BOOKINGS USERS
```

↓
PostgreSQL + PostGIS
↕
Redis
↕
WebSockets
↓
Maps / Payments / Notifications

30. Geolocation Requirements

- Automatic GPS acquisition
- Manual location pin/search
- Location confirmation before request
- Provider distance and ETA calculation
- Real-time provider location during active jobs
- Controlled access to location data
- Avoid unnecessary long-term storage of continuous location history

31. MVP Scope

Customer: authentication, vehicles, emergency request, problem selection, location, matching, booking, status tracking, basic map/tracking, completion, payment record, rating.

Provider: registration, verification, profile, services, online/offline, request acceptance/rejection, job status, completion, earnings.

Admin: login, dashboard, customer/provider management, provider verification, booking monitoring, basic analytics.

32. Phase 2

- Live GPS tracking
- Digital job cards
- Inspection reports
- Estimates and invoices
- UPI payments
- Chat and notifications
- Service history
- Scheduled service
- Garage discovery
- Advanced provider analytics

33. Phase 3

- AI-assisted vehicle diagnosis
- Intelligent provider matching
- Predictive maintenance
- Smart pricing
- Fleet management
- EV specialist network
- Parts marketplace

- Insurance integrations
- Corporate accounts
- Subscription plans

34. AI RoadResQ

Future AI can interpret natural-language symptoms and recommend an assistance category and qualified provider. AI should assist rather than present uncertain mechanical diagnosis as fact.

```
CUSTOMER SYMPTOMS
↓
AI INTERPRETATION
↓
POSSIBLE ISSUE
↓
RECOMMENDED SERVICE
↓
QUALIFIED PROVIDERS
```

35. Business Model

- Transaction commission on completed services
- Provider subscription plans
- Optional featured provider placement with clear labeling
- Fleet/corporate plans
- Garage-management SaaS subscriptions

36. Key Metrics

North Star Metric: Successfully completed assistance jobs per month.

- Successful matching rate
- Average provider response time
- Average ETA
- Cancellation rate
- Repeat customer rate
- Customer rating
- Provider acceptance rate
- Provider completion rate
- Provider retention
- GMV and platform revenue

37. MVP Success Criteria

- Customer can request assistance in approximately one minute
- System identifies a suitable available provider
- Provider accepts the job
- Customer sees job progress
- Provider completes the service digitally
- Platform records invoice/payment information
- Customer can review the provider

38. Security Requirements

- Secure authentication and authorization
- OTP verification
- Role-based access control
- API authorization
- Encrypted communication
- Secure payment processing
- Protected document/file access
- Audit logs
- Rate limiting and input validation
- Secure file uploads

39. Privacy Requirements

RoadResQ may process customer information, vehicle information, phone numbers, location, payment information, provider documents, and service history. The system should follow data-minimization principles and expose information only to parties that need it for the active service workflow.

40. Product Principles

- Emergency First — requesting help must be effortless
- Trust Before Convenience — verified and capable providers matter
- Transparency — users should understand price and process
- Real-Time Visibility — users should know what is happening
- Digital Records — important interactions should remain accessible
- Provider-Friendly — mechanics should gain value, not unnecessary complexity
- Mobile First — roadside assistance is primarily phone-driven

41. RoadResQ + Better Garage

RoadResQ is the customer-facing roadside assistance and automotive-service network: “I need help with my vehicle.”

Better Garage is the garage-facing operating system: “I need to manage my garage.”

The two products can share provider, customer, vehicle, job, billing, and service-history infrastructure while serving different workflows.

42. Future BusinessOS Connection

The broader BusinessOS concept can generalize the garage-management layer into a metadata-driven operating system for multiple business types. RoadResQ can remain the specialized automotive customer/service network.

43. Product Roadmap

```
PHASE 1 — FOUNDATION
Authentication • Customer • Provider • Vehicles
Emergency Request • Matching • Admin

PHASE 2 — SERVICE EXPERIENCE
Live Tracking • Job Cards • Inspection
```

Estimates • Invoices • Payments • Reviews

PHASE 3 — AUTOMOTIVE PLATFORM
Garage Discovery • Scheduled Services
Garage Management • Inventory • Fleet • Analytics

PHASE 4 — INTELLIGENCE
AI Diagnosis • Smart Matching
Predictive Maintenance • Smart Pricing • AI Assistant

PHASE 5 — ECOSYSTEM
Parts Network • Insurance • EV Network
Fleet Partnerships • National Provider Network

44. Final Product Definition

RoadResQ is a location-aware automotive assistance platform connecting vehicle owners with verified mechanics, garages, towing providers, and other automotive service professionals.

It manages the complete assistance journey: Request → Match → Track → Inspect → Approve → Repair → Pay → Review.

Its long-term goal is to become a trusted digital infrastructure layer for India's fragmented automotive service ecosystem.

45. Product Positioning

Primary tagline: Your vehicle breaks down. We bring help.

Alternative: Roadside trouble? RoadResQ.

Positioning: The smarter way to get vehicle assistance.

46. End-to-End Example

Vehicle breaks down
↓
Open RoadResQ
↓
GET HELP NOW
↓
Select vehicle + problem
↓
Confirm location
↓
Provider matching
↓
Provider accepts
↓
Live ETA / tracking
↓
Provider arrives + OTP
↓
Inspection
↓
Estimate / approval
↓
Repair
↓
Invoice
↓
Payment
↓
5-star review
↓

Service saved to vehicle history